



ReMeDi telemedicine setup is used in the remotest rural areas for conducting different types of medical tests

vices on the lowest possible bandwidth with minimal resources. Thus, while solutions in the West might work on 5 MB, in India they need to work at 500 kB or lesser. Cisco's HealthPresence 2.5, for instance, can work even at 300 kbps."

He further says, "The second design consideration is the language support. Cisco's solutions work around this using a special queuing technology that enables patients to find doctors who speak their language and have the right specialisation."

"The third design consideration in emerging countries is that patients sometimes want an unscheduled visit; we could call it *ad hoc* consultations. Cisco caters to these using scheduling systems, and also by bringing in 'presence' within the application to enable paramedics to find any doctor who may be available," Gupta stated.

In the race to overcome these constraints and come up with the best and most cost-efficient solution, the industry has absorbed several interesting technologies from the world of electronics and telecommunications. While some solutions are characterised by advanced technologies, others focus on understanding and catering to rural India.

Let us look at some of the telemedicine and health monitoring solutions being used in India, and what makes them successful.

High-tech in small packages

The extensive use of MEMS is one of the greatest developments in the field

of telemedicine and health monitoring in the last few years.

"MEMS is the technology behind various types of silicon sensors and nano-pumps. STMicroelectronics provides consumer sensor technologies that include multiple-axis motion and orientation sensors, compass, pressure,

temperature, sound (microphone) as well as biological and molecular compounds. These sensors can now be applied to monitor various aspects of the human anatomy and organ functions where accuracy and sensitivity are critical," says Vivek Sharma, regional vice president, Greater China and South Asia Region-India Operations, and director-India Design Centres, STMicroelectronics. ST adds these capabilities to its low-power microcontroller and communication chip technologies, and offers several innovative healthcare products.

ST's Body Gateway (BG), an integrated remote monitoring solution for advanced telemedicine platforms, is an interesting example. It is a lightweight and low-power electronic system for measurement and management of physiological parameters like electrocardiogram (ECG), heart rate, breathing rate, weight, etc. A tiny module that can be taped to the patient's chest, the BG collects data from a suite of sensors, performs initial processing, stores the data and prepares it for transmission over a network to the remote application server.

Another interesting solution is the eye pressure monitor to monitor Glaucoma. Developed by ST in collaboration with Sensimed, the solution is based on a smart contact lens that uses a tiny embedded strain gauge to monitor the curvature of the eye over an extended period of time, providing valuable disease management data that is not currently obtainable using

conventional ophthalmic tests.

Switzerland-based Debiotech and ST have developed silicon-based microfluidic devices that deliver insulin using a miniature disposable insulin pump. Insulin pump therapy is emerging as an attractive alternative to individual insulin injections. The patient is connected to a programmable pump including a storage reservoir, from which insulin is infused into the tissue under the skin, day and night, according to the specific needs of the patient. The system uses miniature MEMS-based technology, and also supports remote administration.

VereMTB is a multiplexed molecular diagnostic lab-on-chip that can identify the specific mycobacterium causing the infection and drug resistance in less than three hours from natural samples. It eliminates the need for culturing—the most time-consuming part of the traditional method. Additionally, due to its compact size, the system can be deployed in a wide range of settings at the point-of-need.

"The remote monitoring solutions from ST are based on the most innovative MEMS devices and detect even the slightest physiological movements. These signals are, in turn, processed by an internal microcontroller, and sent via Bluetooth, Zigbee, Internet, GSM or local network to a control station for monitoring, diagnosis and urgent life-support therapy," says Sharma.

ST's health monitoring solutions use a range of ultra-low power microcontrollers (the STM32L and STM8L series) based on a proprietary 130nm ultra-low-leakage process technology, which decreases overall energy consumption without compromising on speed and functionality. The company also has the required technologies for both short-distance (monitoring-device-to-recording device) communication as well as longer-distance (device-to-hospital) communication. Steady communication is enabled through its STM32W wireless network line of microcontrollers as well as its Bluetooth, NFC, Smartcard and RFID solutions.